

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		award (1) for any of following copper(II) ions are blue copper(II) ions / Cu^{2+} are formed copper(II) nitrate is formed a copper(II) salt/solution is formed award (1) for any of following copper atoms lose electrons / are oxidised $\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$ $\text{Cu} - 2\text{e}^- \rightarrow \text{Cu}^{2+}$ copper displaces the silver copper is more reactive than silver	2			2		
		(ii)		2 $\text{AgNO}_3 + \text{Cu} \longrightarrow \text{Cu(NO}_3)_2 +$ 2 Ag award (1) for product award (1) for balancing only if product correct		2		2	1	
	(b)	(i)		$\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \longrightarrow \text{AgCl}(\text{aq})$ $\text{Ag}^+(\text{s}) + \text{Cl}^-(\text{s}) \longrightarrow \text{AgCl}(\text{s})$ $\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \longrightarrow \text{AgCl}(\text{s})$ ✓ $\text{Ag}^+(\text{s}) + \text{Cl}^-(\text{s}) \longrightarrow \text{AgCl}(\text{aq})$	1			1		